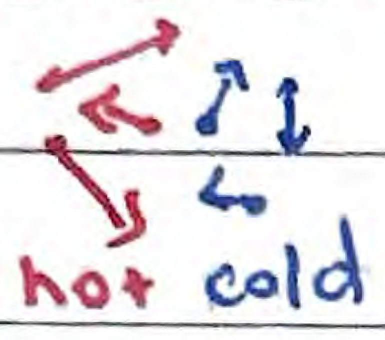
Date. NO.

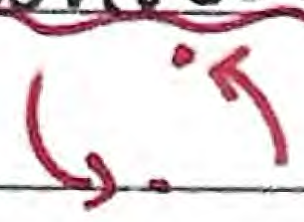
Energy Emission

Recall energy transfer occurs from

1. Conduction – transfer of energy through collisions between neighboring atoms or molecules



2. Convection – transfer of energy by movement of a heated fluid in a cyclical manner



↳ Via molecular flow

3. Radiation – transfer of energy through electromagnetic waves from a hot body to a colder one

Energy commonly refers to that of mechanical, thermal, electric, etc.

Rate of energy transfer is given by power, $\boxed{P = \frac{\Delta E}{\Delta t}}$

Intensity – Amount of energy passing through a unit area per unit time, or power per unit area

↳ Rate of energy transfer perpendicular to the direction of propagation

$$\boxed{I = \frac{P}{A} = \frac{E}{A \cdot t}}$$

for...

I - intensity
P - power
A - area
E - energy
t - time

given in $W \cdot m^{-2}$

Blackbody Radiation

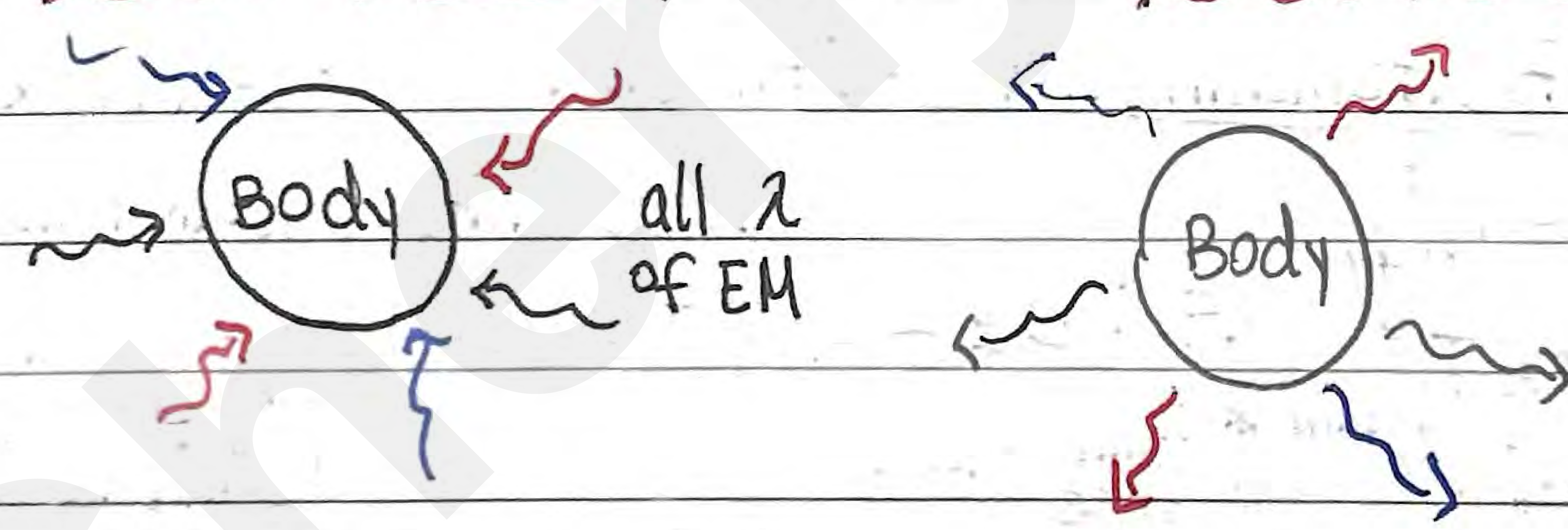
Refers to the thermal radiation emitted by all bodies

↳ Can be infrared, or visible, etc. depending on temperature.

A perfect black body, one that absorbs/emits blackbody radiation, is defined as...

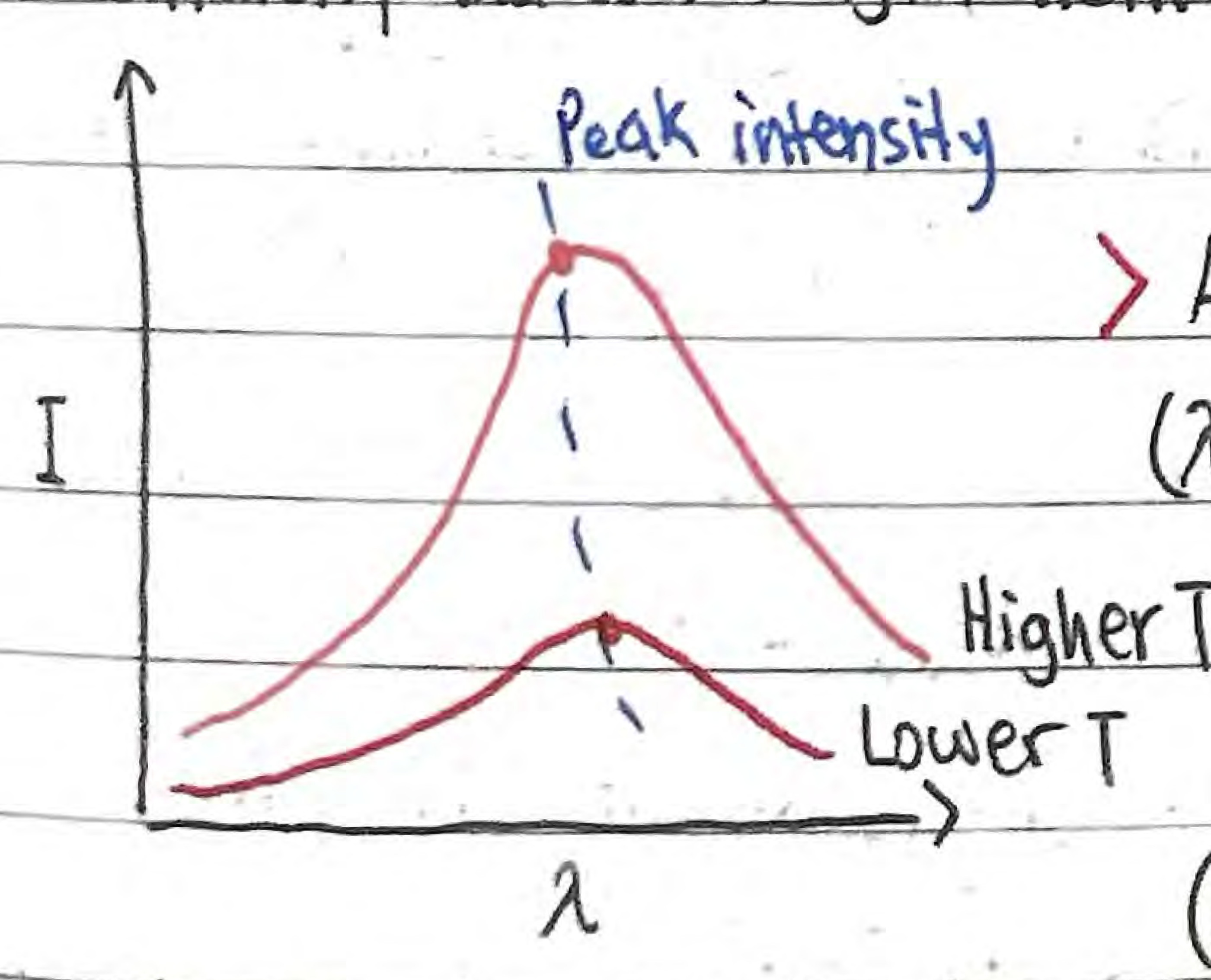
⌈ An object that absorbs all of the radiation incident on it and does not reflect nor transmit any ⌋

It's a perfect absorber, hence also perfect emitter



Absorption of all λ makes body appear black

Intensity and wavelength distribution of any emitted wave depend on temp.



> As $T\uparrow$, $I_{peak}\uparrow$. But $\lambda_{peak}\downarrow$
(λ shift left)

↳ This indicates temperature causes a change in color (as λ with highest intensity is perceived)

Blackbody Radiation

Apparent Brightness - Intensity of radiation on Earth received from a star

↳ Depends on how much light the star emits, how far the star is

↳ Tells us brightness as measured by observer

Luminosity - Total power output of radiation emitted by a star

↳ Total power output of a star, brightness at its surface

Hence, apparent brightness (intensity) is typically a fraction of luminosity

Stefan-Boltzmann Law - Total radiated power of a blackbody depends on its absolute temp. and surface area

$P_{emit} = \sigma A T_{body}^4$

$P_{absorb} = \sigma A T_{surround}^4$ ⇐ $P = \sigma A T^4$

$P_{net} = e \sigma A (\Delta T)^4$

↳ when object is not in equilibrium (temp. in change)

$\boxed{\text{where...}}$

P - Total power absorbed/emitted across all wavelengths; A - surface area of body

σ - Stefan-Boltzmann constant ($5.67 \times 10^{-8} W m^{-2} K^{-4}$)

T - Absolute temperature of surrounding/body → $e = \frac{\text{power radiated by object}}{\text{power emitted by black body}}$

useful since ↓ planets, unlike stars, aren't approximate to black bodies

no emission (e.g. black hole) ↓ perfect black body (e.g. star)

L = luminosity $= P = e \sigma A T^4$ for $e \in (0, 1)$, emissivity of surface

Wien's displacement law - relates observed λ to surface temperature

$$\lambda_{max} \cdot T = 2.9 \times 10^{-3} mK$$

for $\lambda_{max} \propto \frac{1}{T}$

$\boxed{\text{where}}$ λ_{max} - greatest intensity λ

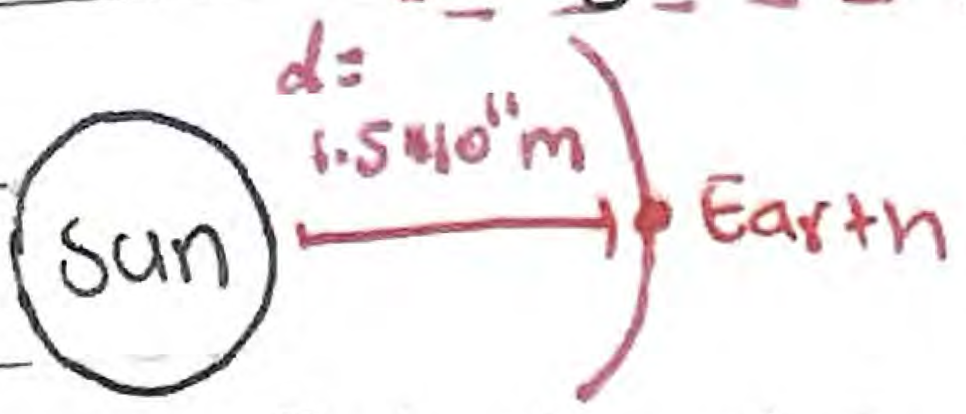
T - surface temperature

↳ As temp. ↑, λ emitted ↓

Energy Balance Model of the Planet

Solar Constant - Intensity of Sun's radiation arriving perpendicularly to the Earth's position at average orbit radius

↳ $S_{avg} = 1.36 \times 10^3 \ W m^{-2}$ (but changes because of elliptical orbit)



Sun's luminosity, $L_\odot$, spread over all sides (at Earth's orbit, total spread area is $4\pi d^2$)

$S = \frac{L_\odot}{4\pi d^2}$ ↳ intensity per m^2

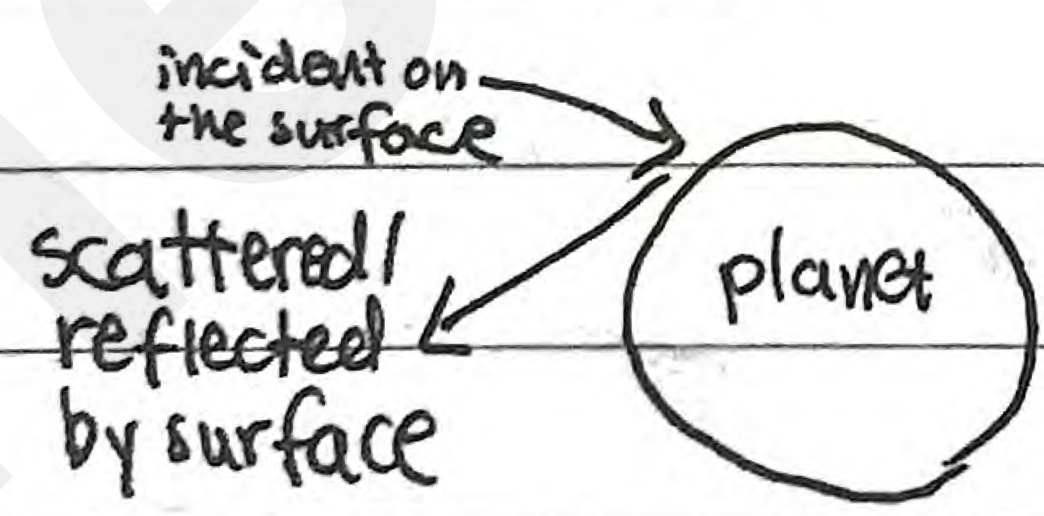
For Earth, it captures intensity over circular cross section, spreads it over entire surface : $S \times \pi r_{earth}^2 \div 4\pi r_{earth}^2 \Rightarrow \frac{S}{4}$ (mean solar intensity at planet surface)

*But we need to quantify ability of a non-blackbody to: $\begin{cases} \text{absorb, reflect, scatter thermal radiation} \\ \text{emit thermal radiation} \end{cases}$

$$> \alpha = \text{albedo} = \frac{\text{total power scattered}}{\text{total power incident}} = \frac{\text{total intensity scattered}}{\text{total intensity incident}}$$

consequently... $\alpha = 1 - \frac{\text{total absorbed}}{\text{total incident}}$

where...



*Scatter ≠ emission
⇓ reflected off
⇓ acts as source
(scatter happens before absorbed)

$\boxed{0} \le \alpha \le \boxed{1}$ when all radiation is scattered

when all power is absorbed

E.g.

Material	Albedo
ocean water	0.1
forest	0.1
ocean ice	0.4
......	...

Energy Balance Model of the Planet

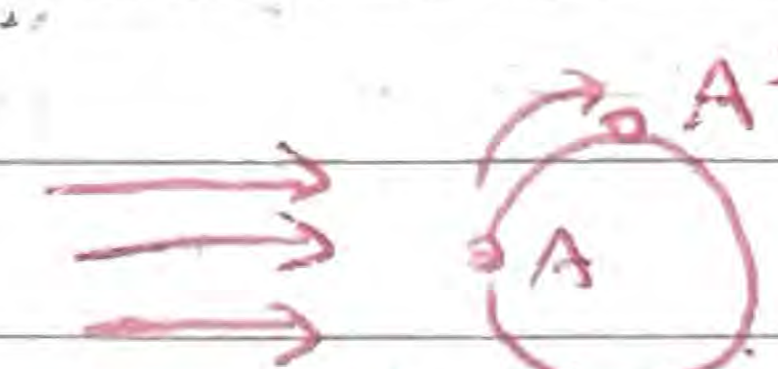
E.g. Radiation of intensity $610\ \text{W m}^{-2}$ was incident perpendicular on the surface of a lake

a) If the lake has an albedo of 0.18, calculate how much intensity was absorbed by the surface

$$I_{absorbed} = 610(1-\alpha)$$
$$= 610(1-0.18) = \underline{500\ \text{W m}^{-2}}$$

b) Describe how answer in (a) would change later in the day

$I_{absorbed}$ decreases :



distance ↑, so I ↓

Model assumption:

- For a planet with no atmosphere in thermal equilibrium the outgoing radiated intensity equals the absorbed intensity

Energy Balance Model of the Planet

The IB uses the following:

actually received

all received is emitted

> Radiant thermal energy received by planet (with albedo considered) = Radiant thermal energy emitted by planet

this is also energy absorbed by planet

both under same time interval

or...

$$\boxed{\frac{S}{4}(1-\alpha) = e\sigma T^4}$$

emissivity (with respect to black body)

albedo (with respect to body)

In the case of the Earth...

$$\frac{S}{4}(1-\alpha) = e\sigma T^4$$

$$\frac{1.36\times10^3}{4}(1-0.315) = (0.61)(5.67\times10^{-8})T^4$$

$$T = 286\,K = 13°C$$

← Average temperature of the Earth.

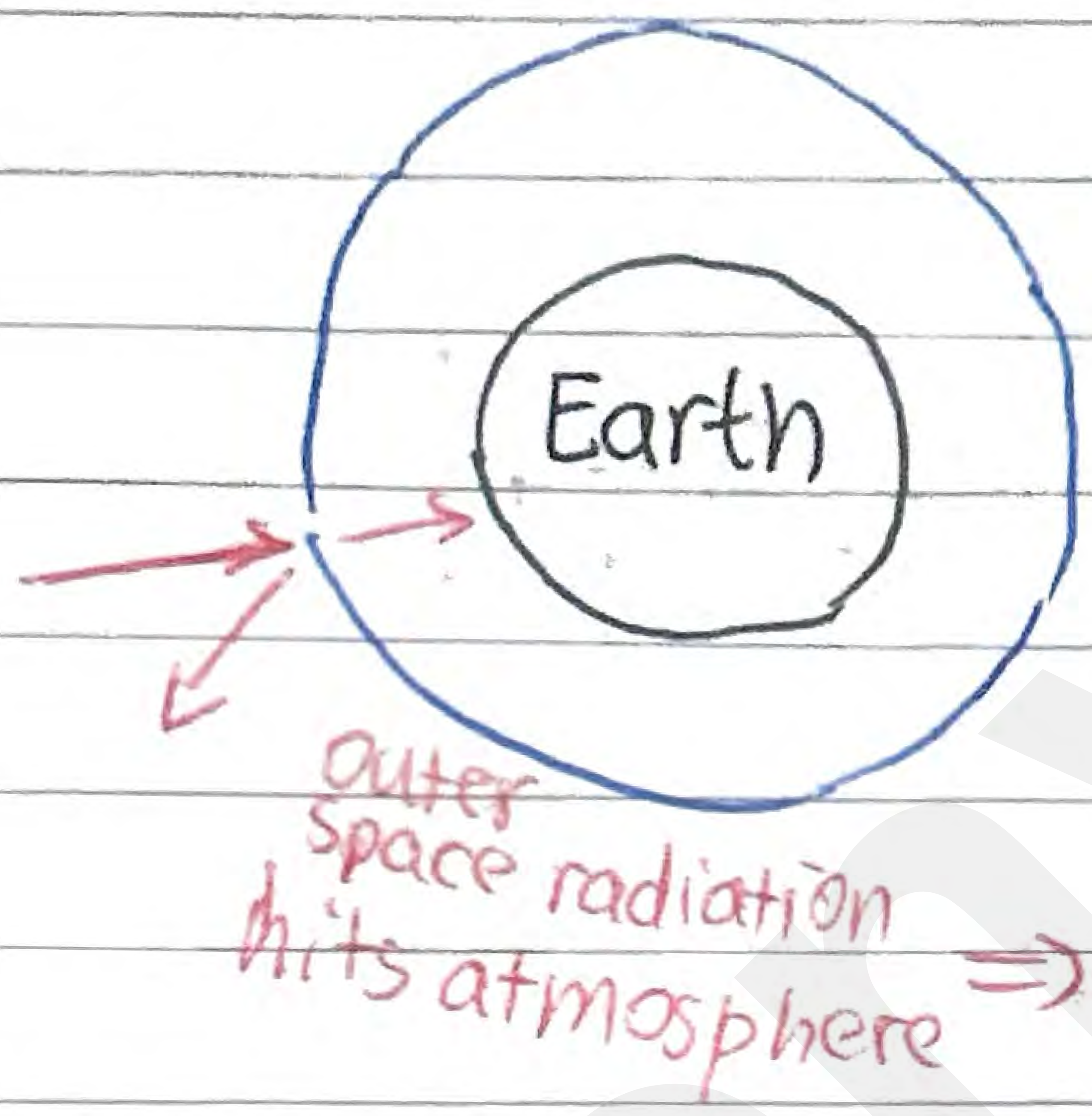
Although model is highly simplified, it's reasonably accurate in concluding that:

1. The values of emissivity and albedo have a fundamental effect on the planet's temperature
2. The values of e and α are being changed by human activity
3. Small changes in the Earth's temperature may produce large changes in the Earth's climate.
   - ↳ Such changes are harmful to our lives

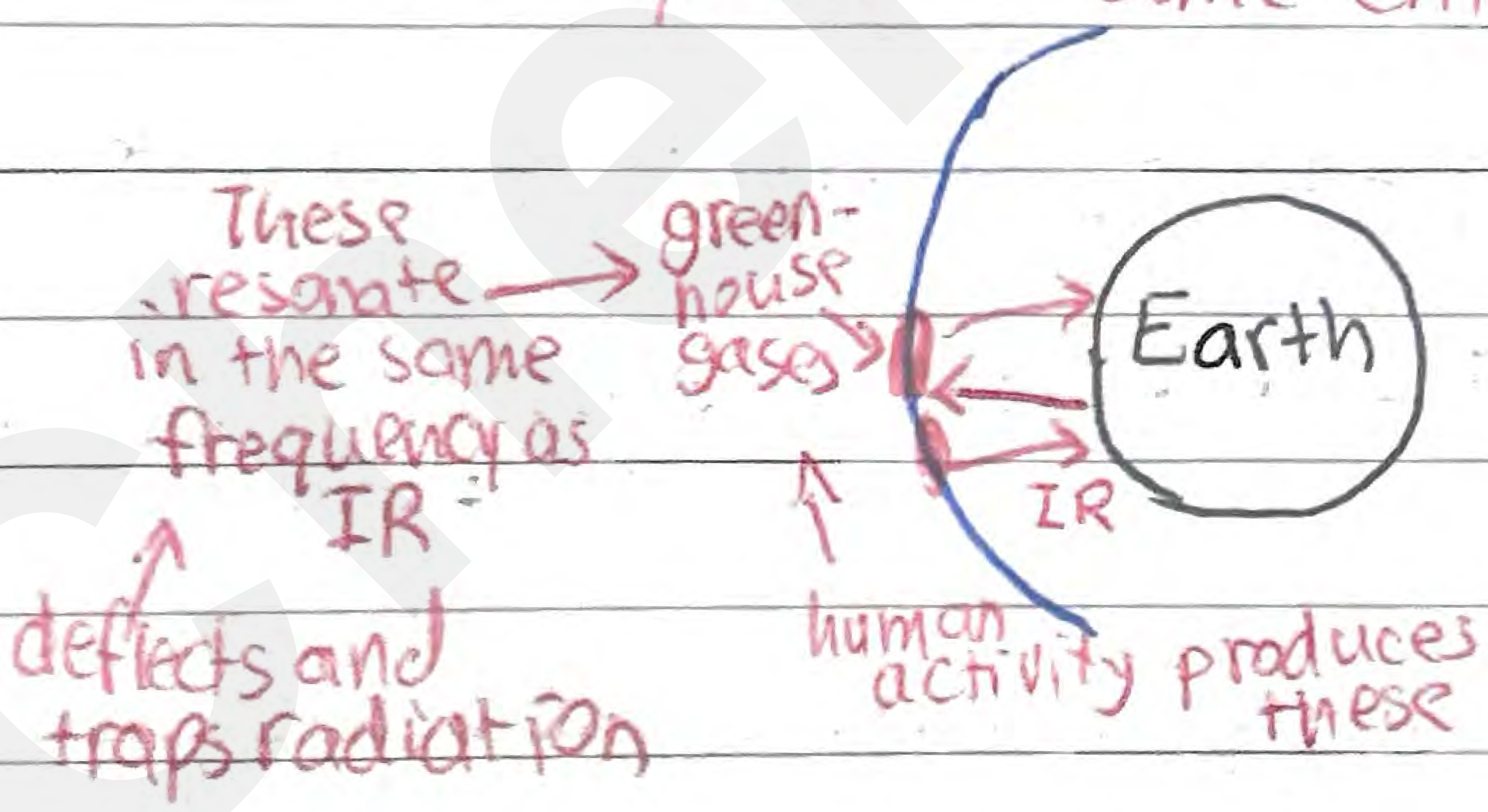
Greenhouse Effect

IB's model reveals that human activity causes temperature rises on the Earth's surface

· Greenhouse effect furthers this idea (adds to simplified model) since emission can lead to more absorption from reradiated energy)



⇒ some are deflected, some enter



Greenhouse Effect – Physical mechanism by which some of the radiation (IR) emitted by the surface of the planet is absorbed by the greenhouse gases in the atmosphere and re-radiated towards the surface

Greenhouse Effect

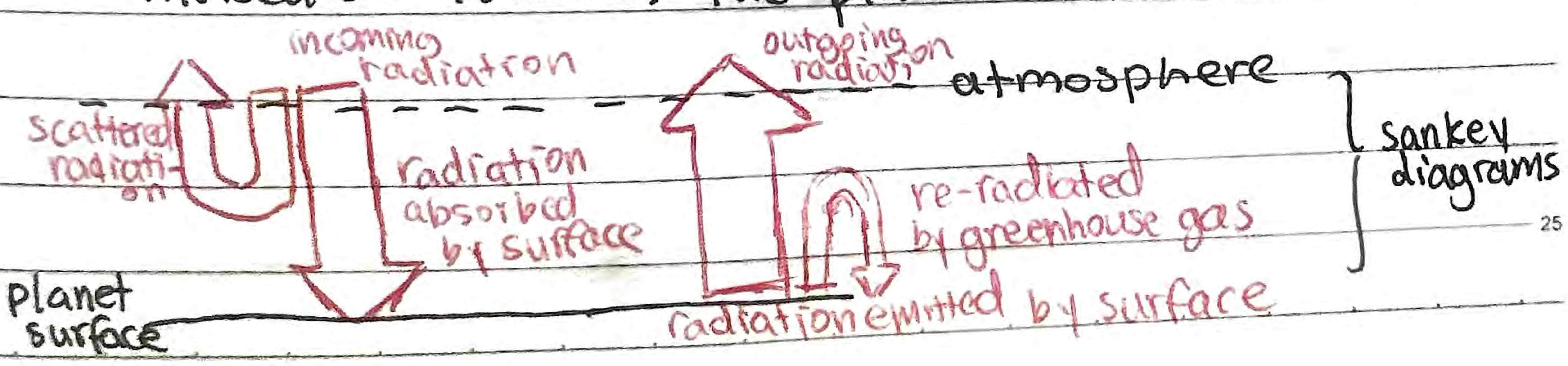
Essentially, gases in our atmosphere trap energy and release it into the atmosphere
↳ This causes atmosphere to warm

Main Greenhouse Gases include...

sources:	natural	human
› Methane	wetlands	agriculture
› water vapor	clouds, oceans	fossil fuel combustion
› carbon dioxide	plant respiration, surface ocean	fossil fuel combustion
› nitrous oxide	forests, oceans	agriculture

Main steps

1. Absorption of radiation causes resonance with greenhouse gas molecules (radiation matches energy difference between molecular levels)
2. Emission of radiation from greenhouse gas molecules towards the planet surface

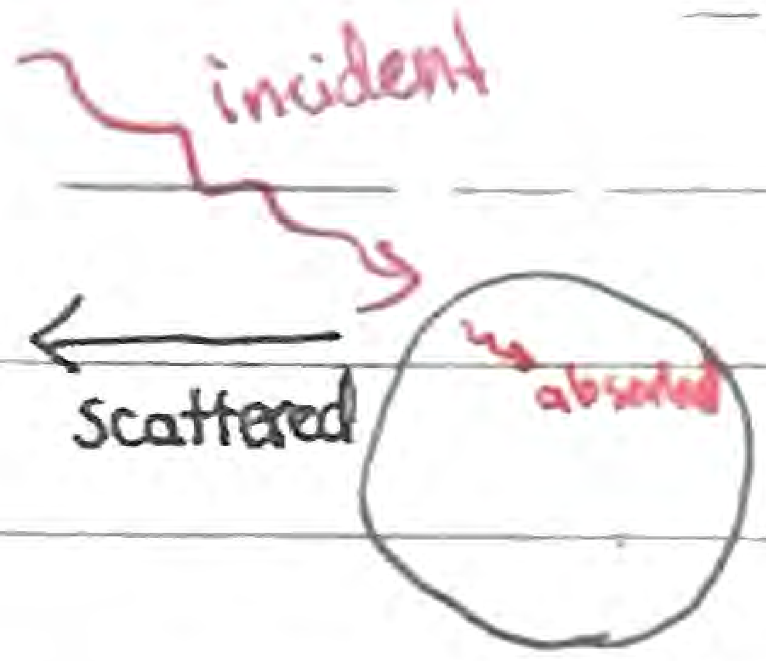


Date NO.

Greenhouse Effect

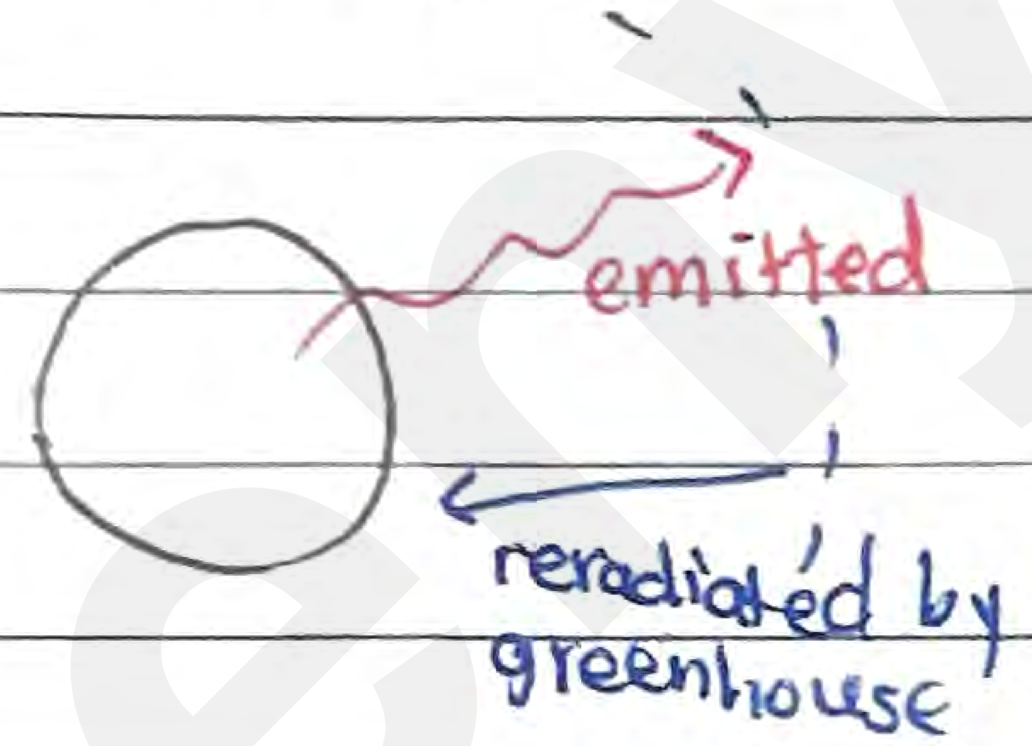
Conservation of Energy

All absorbed must be emitted



albedo considered

But then, the absorbed must be reradiated



← This is absorbed by Earth too

Hence...

Absorbed from incident + Reradiated by greenhouse = emitted by earth